Hao Lee

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Professional Summary

MS in Mechanical Engineering (Data Science) at University of Washington, specializing in robot learning, and multi-agent systems. Hands-on experience building end-to-end VLA pipelines, GNN-based drone formation control, and optimal control for robotic manipulation. Seeking a Research Assistant position to contribute to real-world robot learning and embodied AI research.

Technical Skills

Languages, Frameworks & Tools: Python, C++, MATLAB ,ROS2, PyTorch, CasADi/IPOPT, MuJoCo, CUDA, Docker, Git
Robotics & Control: Robot Manipulation, MPC, Path Planning, SLAM, Behavior Trees, FK/IK, Multi-Agent Coordination
ML / AI: Deep RL, GNN, Transformer, World Models (RSSM), Imitation Learning, Multimodal LLM, VLA

Education

University of Washington — **Sep 25 –Present**
Master of Science in Mechanical Engineering (data science track) — *Seattle, U.S.A*
Related coursework: CSE546 Machine learning / CSE571 AI-robotics / CSE579 Reinforcement Learning / EE546 control sys theory

National Cheng Kung University — **Sep 21 – Jun 24**
International bachelor's degree of Energy Engineering under dept. of Aeronautical Engineering — *Tainan, Taiwan*

Research Experience

LLM-Inferred Reward Functions for Task-based RL — **Apr 26 – Present**
Capstone Project, UW (CSE579 Reinforcement Learning) — *Seattle, U.S.A*
- Built an RLHF-style reward learning pipeline that replaces human annotators with an LLM: given environment source code, an LLM generates both a fixed reward function r_{fixed} and a semantic trajectory summarizer via RAG
- Trained an ensemble reward model (3× MLP, Bradley-Terry loss) with active learning via disagreement scoring to minimize LLM queries; combined reward $r_{total} = r_{fixed} + g \cdot R_{\varphi}$ used to train PPO on HalfCheetah-v5

Vision-Language-Action (VLA) Robot Control Pipeline — **Jan 26 – Mar 26**
Capstone Project, UW (CSE571 AI-Robotics) — *Seattle, U.S.A*
- Built an end-to-end 4-stage VLA pipeline: natural-language command → semantic waypoint planning → 3D projection → Cartesian trajectory execution on a 6-DOF robot arm
- Integrated Google Gemini as vision-language planner to infer semantic 2D waypoints from RGB frames, then lifted pixel-space waypoints into 3D robot-frame coordinates via stereo depth projection with spatial-temporal median filtering

Multi-Drone Autonomous Minefield Mapping & Path Planning — **Jan 26 – Present**
Washington Aerial Robotics Team (IARC Mission 10)) — *Seattle, U.S.A*
- Architected peer-to-peer coordination layer using ROS2/FastDDS with gossip-protocol-style diffusion across 4 autonomous agents, achieving sub-100ms propagation latency
- Designed decentralized task-auction mechanism enabling agents to independently estimate costs and compete via claim broadcasts and implemented hierarchical state machine with priority-ordered behavior tree for reactive collision avoidance

GNN Formation Control for Multi-Drone Systems — **Jan 26 – Mar 26**
Capstone Project, UW (EE546 Advanced Control Theory) — *Seattle, U.S.A*
- Designed a GNN policy trained via imitation learning to maintain chain formation across leader-follower drone networks
- Achieved zero-shot generalization from n=4 to n=8 followers using permutation-equivariant shared weights

World Model Comparison Study (Dream to Control) — **Jan 26 – Mar 26**
Capstone Project, UW (ME571 Data Driven Control) — *Seattle, U.S.A*
- Extended and evaluated DreamerV1 to compare RSSM, LinearSSM, and MLP dynamics models on Pendulum environment

Deep Reinforcement Learning for Quadrotor Flight Control — Undergraduate Thesis — **Sep 23 – Jun 24**
NCKU Bioinspiration Lab (Dr. Chen) — *Tainan, Taiwan*
- Implemented DDPG algorithm in simulation to train an autonomous quadcopter flight controller, targeting higher stability and faster convergence than classical PID controllers
- Performed ablation studies on reward shaping, action-space normalization, and noise injection to optimize flight policy; presented findings as pioneering exploration of DRL-based UAV control within the lab group

Work Experience

Data Engineer Intern — **Jun 24 – Aug 24**
AUO-Digitech — *Hsinchu, Taiwan*
- Processed and curated large-scale manufacturing datasets to build training pipelines for predictive ML models; performed hyperparameter tuning and performance evaluation using PyTorch-based frameworks. Developed NLP applications (chatbots, text classification) leveraging fine-tuned BERT and GPT models
- Supported cross-functional R&D on drone applications in factory automation, bridging ML knowledge with robotics system integration